Equity Performance Project

Analysis of Investor Personas and Index Tracking

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1 Problem and Model Formulation

The main objective of this project is to analyze the performance of a specific equity portfolio from the perspectives of three different hypothetical personas. This analysis will be extended to include a formal evaluation of risk-adjusted returns and the construction of an index-tracking portfolio.

1.1 Core problem definition

The core analysis is centered on a portfolio composed of 10 U.S. stocks, which were selected to represent a mix of technology, dividend aristocrat[1], and high-dividend yield sectors. The portfolio is defined by

- **Technology:** MSFT, AAPL, IBM
- **Dividend aristocrat:** XOM, WTM, KO, JNJ
- **High dividend yield:** VZ, KHC, USB

The performance of this 10-stock target portfolio will be evaluated using historical market data, we assume a specific weighting strategy defined in Section 2.

1.2 Persona models (Stochastic Models)

The performance of the target portfolio is evaluated through three personas, each representing a different investment horizon and objective. The PnL for each is modeled as a stochastic process dependent on the historical price path of the portfolio's assets.

1.2.1 Buy & Hold (BH) model

This persona represents a long-term investor who acquires the portfolio at the beginning of the analysis period and holds it until the end.

- **Objective:** Maximize total return over the entire period, inclusive of all cash dividends.
- **Model:** The PnL is a single-stage calculation: $\mathrm{PnL}_{\mathrm{BH}} = \mathrm{Value}_E - \mathrm{Value}_S$

[1] A dividend aristocrat is an S&P 500 company that has increased its dividend annually for at least 25 years, demonstrating a commitment to returning cash to investors

1.2.2 Hedge Fund (HF) model

This persona represents an institutional investor that marks-to-market[2] and captures PnL changes on a daily basis.

- **Objective:** Generate consistent, positive daily returns (alpha) while managing daily volatility.
- **Model:** The PnL is a multi-stage stochastic series: $\text{PnL}_{HF}(t) = \text{Value}_t - \text{Value}_{t-1}$ for each day t.

1.2.3 Day Trader (DT) model

This persona represents a short-term speculator who seeks to profit from intraday price movements and holds no overnight positions.

- **Objective:** Achieve positive PnL from a defined intraday trading strategy.
- **Model:** The PnL is a strategy-dependent daily series, $\text{PnL}_{DT}(t) = PnL_{\text{Strategy},t}$.

1.3 Risk Constraints and Performance Metrics

A simple return analysis is insufficient. We introduce risk constraints by evaluating each persona's PnL stream using standard performance metrics.

- **Sharpe Ratio:** Measures risk-adjusted return, defined as:

  $$\text{Sharpe Ratio} = \frac{E[R_p - R_f]}{\sigma_p} \tag{1}$$

  where R_p is the portfolio's return, R_f is the risk-free rate, and σ_p is the standard deviation (volatility) of the portfolio's excess return.

- **Value at Risk (VaR):** Defines the maximum expected loss at a given confidence level $1-\alpha$. It is the $1-\alpha$ quantile of the PnL distribution, $F_X(x)$.

  $$\text{VaR}_\alpha(X) = -\inf\{x \in \mathbb{R} : F_X(x) > \alpha\} \tag{2}$$

  For this project, we will use the non-parametric Historical Simulation method.

[2] Accounting method that values assets and liabilities at their current market price rather than their historical or original cost

- **Conditional Value at Risk (CVaR):** Measures the expected loss *given* that the loss exceeds the VaR.

$$\mathrm{CVaR}_\alpha(X) = E[-X \mid -X \geq \mathrm{VaR}_\alpha(X)] \tag{3}$$

1.4 Index Tracking Model (Optimization Model)

This task involves creating a replicating portfolio from a universe of other stocks (e.g., S&P 500 components excluding the 10 target stocks) to mimic the returns of our target portfolio. This is formulated as a constrained optimization problem.

- **Objective Function:** Minimize the Tracking Error Variance (TEV), which is the variance of the difference between the target portfolio's returns (R_T) and the replicating portfolio's returns (R_R).

$$\min_{w} \quad \mathrm{TEV} = E\left[(R_T(t) - R_R(t))^2\right] \tag{4}$$

  where $R_R(t) = \sum_{i=1}^{N} w_i R_i(t)$ and w_i is the weight of stock i in the replicating portfolio.

- **Constraints:** The optimization is subject to several constraints:

  1. **Full Investment:** Weights must sum to one.

$$\sum_{i=1}^{N} w_i = 1 \tag{5}$$

  2. **Long-Only:** No short selling is permitted.

$$w_i \geq 0 \quad \forall i \in \{1, \ldots, N\} \tag{6}$$

  3. **Cardinality Constraint:** The number of stocks in the replicating portfolio must not exceed k (e.g., $k = 20$). This is represented using the L_0-norm:

$$\|w\|_0 \leq k \tag{7}$$

  4. **Buy-In Constraint:** Any selected stock must have a minimum weight of l (e.g., $l = 1\%$). This makes the weights semi-continuous:

$$w_i \in \{0\} \cup [l, 1] \quad \forall i \tag{8}$$

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A Python Code

```
# Paste your Python code here
import yfinance as yf
import pandas as pd

# ...
```

B Data Tables